

day 10 prep

integration

1. rectangle rule
2. trapezoid rule
3. think through the differences in how you code this up

integrating $\sin(x)$ from 0 to π .

this should be $-\cos(x)|_0^\pi = -\cos(\pi) + \cos(0) = 2$

```
In [17]: import numpy as np
import matplotlib.pyplot as plt
```

```
In [20]: -np.cos(np.pi) + np.cos(0)
```

```
Out[20]: 2.0
```

```
In [48]: x = np.linspace(0, 5, 10)
y = x**2 + x + 1
```

```
In [49]: # need a dx
dx = x[1]-x[0]
```

```
In [50]: ydx = y * dx
```

```
In [51]: y
```

```
Out[51]: array([ 1.          ,  1.86419753,  3.34567901,  5.44444444,  8.16049383,
                11.49382716, 15.44444444, 20.01234568, 25.19753086, 31.          ])
```

```
In [52]: ydx
```

```
Out[52]: array([ 0.55555556,  1.03566529,  1.85871056,  3.02469136,  4.53360768,
                6.38545953,  8.58024691, 11.11796982, 13.99862826, 17.22222222])
```

```
In [53]: ydx.sum()
```

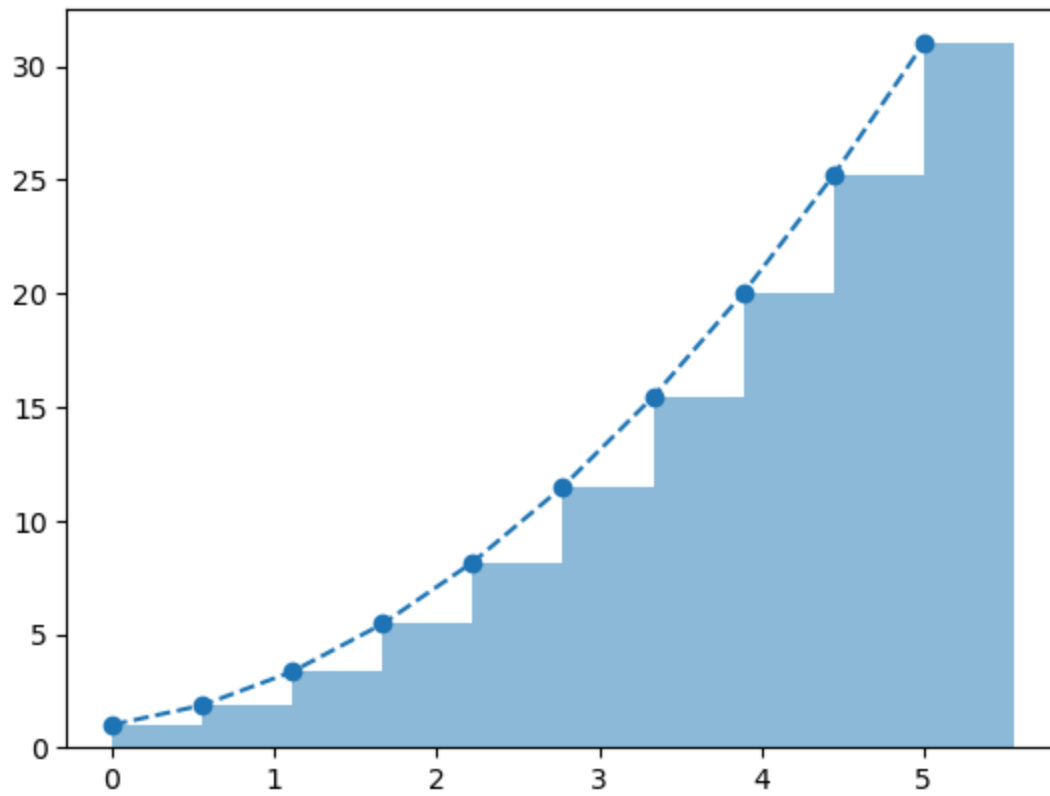
```
Out[53]: 68.3127572016461
```

so that is not too bad when the answer was 2... but we can do better. we need to look at how exactly we were doing this sum.

```
In [54]: fig, ax = plt.subplots()

ax.plot(x, y, 'o--')
ax.bar(x, y, alpha=0.5, align='edge', width = dx)
```

Out[54]: <BarContainer object of 10 artists>



In []:

In []: